

Exercise 2 - Sagelva Klima

Task 1 - Annual precipitation and mean temperature

1.1 Methodology

Daily precipitation and air temperature data from the Svarttjønnbekken station were used to calculate annual precipitation totals and annual mean temperatures for the period 1973–2021. Missing values, represented by –9999 in the dataset, were treated as missing observations and excluded from the calculations.

Annual precipitation was calculated as the sum of the available daily precipitation measurements:

$$P_{annual} = \sum_{i=1}^n P_i$$

where P_i represents daily precipitation and n is the number of available observations during the year.

Annual mean temperature was calculated as the arithmetic mean of the available daily temperature observations:

$$T_{annual} = \frac{1}{n} \sum_{i=1}^n T_i$$

The precipitation record contains no data for the period 2005–2016. Consequently, these years were excluded from the precipitation analysis and are represented as a gap in the corresponding time series. The temperature record contains no data for 2005. In addition, 1993 contains 358 daily precipitation observations rather than a complete annual record and should therefore be interpreted with some caution.

1.2 Annual Precipitation

The calculated annual precipitation shows substantial interannual variability throughout the available record. The mean annual precipitation for the available years between 1973 and 2021 is 1302.01 mm/year. The median annual precipitation is 1287.10 mm/year, while the standard deviation is 268.05 mm/year.

The lowest annual precipitation was recorded in 1974, with 841.50 mm, whereas the highest occurred in 2021, with 1840.75 mm. Thus, the observed range is 999.25 mm, highlighting the considerable variability in annual precipitation between years.

The main precipitation statistics are summarised in Table 1.

Table 1. Summary statistics for annual precipitation at Svarttjønnbekken.

Statistic	Value
Mean	1302.01 mm/year
Median	1287.10 mm/year
Standard deviation	268.05 mm/year
Minimum	841.50 mm/year (1974)
Maximum	1840.75 mm/year (2021)
Range	999.25 mm/year

Several particularly wet years can be identified, including 1973 (1679.3 mm), 1975 (1666.2 mm), 1983 (1655.5 mm), 1992 (1810.3 mm), and 2021 (1840.75 mm). In contrast, relatively dry years include 1974 (841.5 mm), 1977 (906.0 mm), 1996 (893.8 mm), and 1984 (950.2 mm).

The temporal variation in annual precipitation is illustrated in Figure 1, where the substantial year-to-year variability can be observed directly. The horizontal reference line represents the mean annual precipitation calculated from the available years.

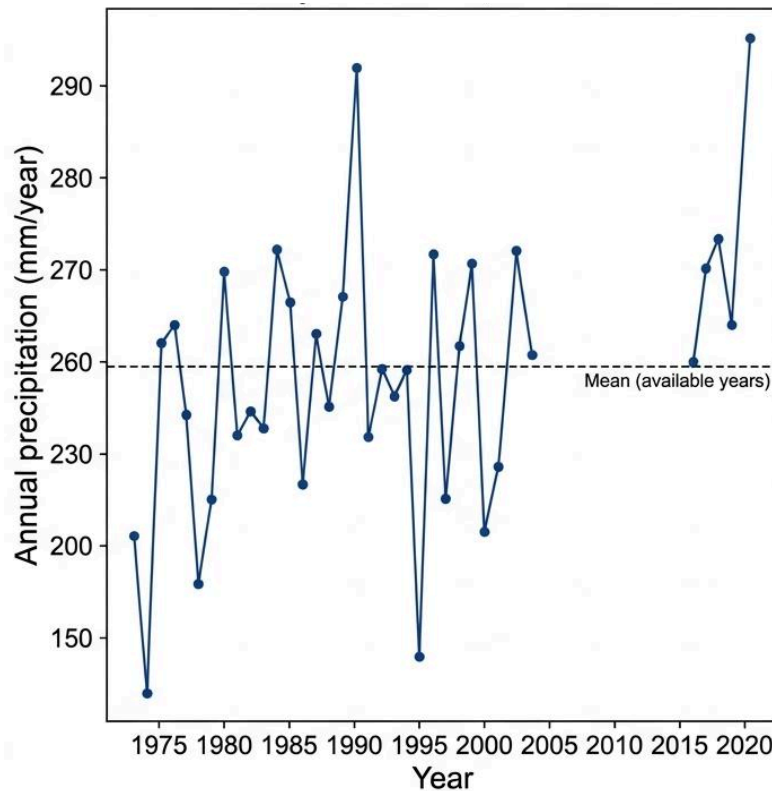


Figure 1. Annual precipitation at Svarttjønnbekken for the period 1973–2021. Annual totals were calculated from the available daily observations. The horizontal reference line represents the mean annual precipitation of 1302.01 mm/year calculated from the available years. The gap between 2005 and 2016 represents missing precipitation data and has not been interpolated.

A major limitation of the precipitation record is the absence of observations between 2005 and 2016. Therefore, this period cannot be used to assess annual precipitation variability or trends. The value calculated for 1993 should also be interpreted with some caution because only 358 daily observations are available for that year.

1.3 Annual Mean Temperature

The annual mean temperature over the available period is 4.08 °C. The lowest annual mean temperature was recorded in 2010, at 1.96 °C, while the highest occurred in 2000, at 5.87 °C.

The main temperature statistics are summarised in Table 2.

Table 2. Summary statistics for annual mean temperature at Svarttjønnbekken.

Statistic	Value
Mean	4.08 °C
Minimum	1.96 °C (2010)
Maximum	5.87 °C (2000)

The annual mean temperature series is presented in Figure 2. The figure shows considerable interannual variability, with generally higher annual mean temperatures occurring during the late 1990s and early 2000s. Particularly high values were recorded in 2000 (5.87 °C), 2002 (5.86 °C), 2003 (5.79 °C), and 2004 (5.67 °C). In contrast, 2010 was notably cold, with an annual mean temperature of only 1.96 °C.

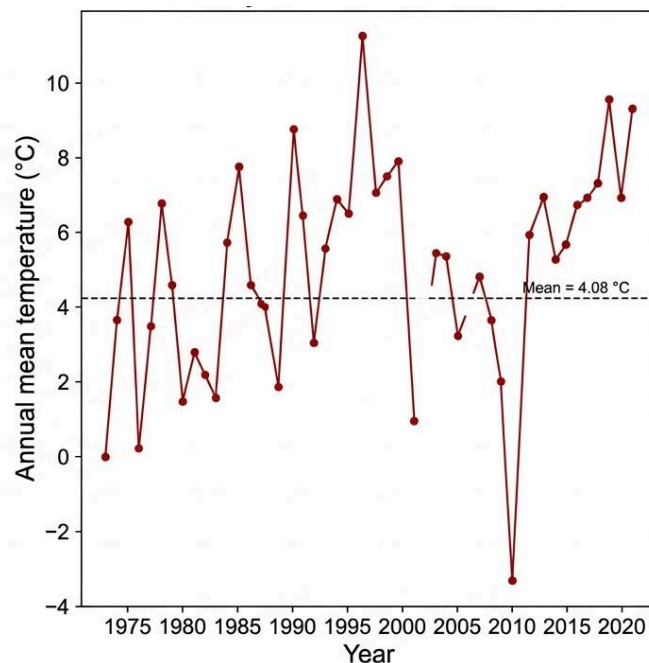


Figure 2. Annual mean air temperature at Svarttjønnbekken for the period 1973–2021. Annual means were calculated from the available daily observations. The horizontal reference line represents the mean annual temperature of 4.08 °C. The gap in 2005 represents missing temperature data and has not been interpolated.

Unlike the precipitation series, the temperature record contains data for almost the entire period, with 2005 being the only missing year. The temperature record therefore provides a more continuous basis for assessing interannual temperature variability over the study period.

Task 2 - Flood Frequency Distribution

2.1 Flood Frequency Distribution

2a. Relative and Cumulative Frequency Distributions

The annual maximum discharge values obtained in Exercise 1 were used to construct a flood series for the period 1973–2025. A total of 53 annual maximum discharge observations were available. The data were grouped into discharge intervals with a bin size of $0.1 \text{ m}^3/\text{s}$, as specified in the exercise.

For each interval, the relative frequency was calculated as the number of observations within the interval divided by the total number of observations:

$$f_i = \frac{n_i}{N}$$

where n_i is the number of annual maximum discharges within a given interval and $N = 53$ is the total number of observations.

The cumulative frequency was then calculated by progressively summing the relative frequencies:

$$F_i = \sum_{j=1}^i f_j$$

The annual maximum discharge ranges from approximately $0.38 \text{ m}^3/\text{s}$ to $2.70 \text{ m}^3/\text{s}$, with a mean value of approximately $1.13 \text{ m}^3/\text{s}$ and a standard deviation of $0.46 \text{ m}^3/\text{s}$. The resulting distributions show that most annual maximum discharges are concentrated around the central part of the distribution, while a smaller number of years exhibit substantially higher peak flows.

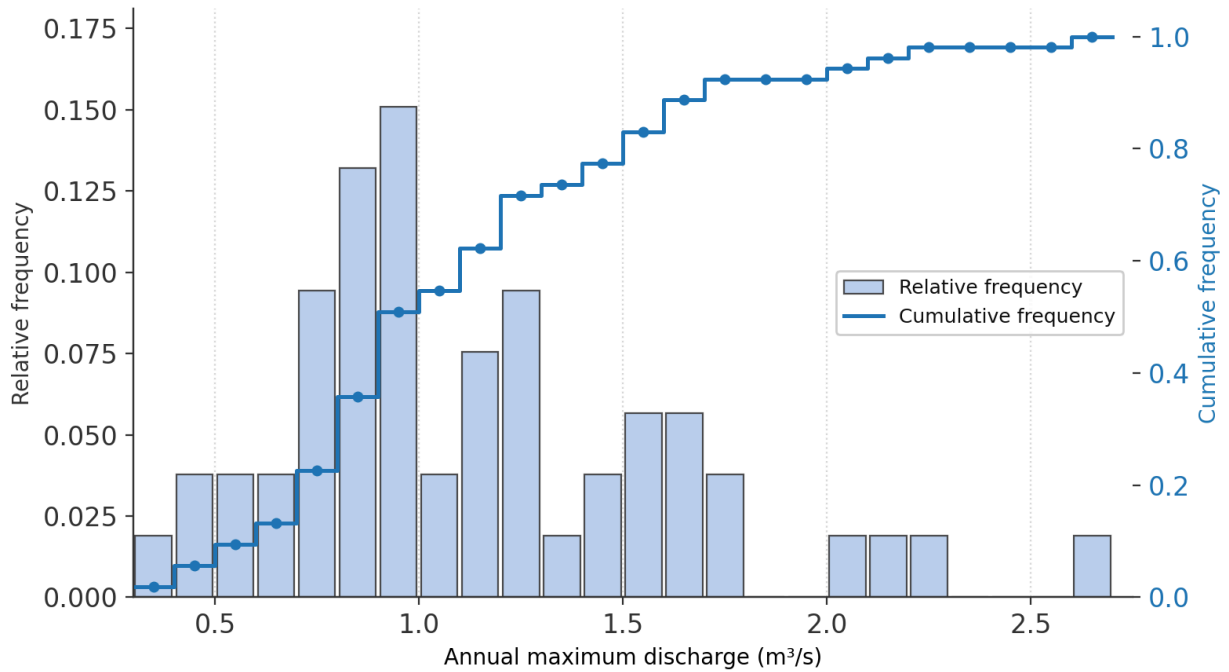


Figure 3. Relative and cumulative frequency distributions of annual maximum discharge at Svarttjønnbekken. The annual maximum discharge data from 1973–2025 were grouped using discharge intervals of $0.1 \text{ m}^3/\text{s}$.

2b. Fitting a Standard Normal Distribution

A standard normal distribution was fitted to the observed flood series following the procedure suggested in the exercise. The mean and standard deviation of the annual maximum discharge series were used to standardise each bin value. For each bin, the upper limit of the interval was used as x , following the example given in the exercise.

The standardised value was calculated as:

$$z = \frac{x - \mu}{\sigma}$$

where $\mu = 1.129 \text{ m}^3/\text{s}$ is the mean annual maximum discharge and $\sigma = 0.459 \text{ m}^3/\text{s}$ is the standard deviation.

The cumulative probability $F(z)$ was then obtained from the standard normal distribution. The theoretical relative frequency for each discharge interval was calculated by subtracting consecutive cumulative probabilities:

$$f_i = F(z_i) - F(z_{i-1})$$

This provides the expected frequency of each $0.1 \text{ m}^3/\text{s}$ discharge interval according to the fitted normal distribution, which can then be compared with the observed frequency distribution.

The normal distribution provides a reasonable approximation of the central part of the flood series, where the observed and theoretical frequencies show relatively similar behaviour. However, the agreement becomes poorer towards the upper end of the distribution. The annual maximum discharge series is positively skewed, with a skewness of approximately 1.44, indicating a longer upper tail than would be expected from a normal distribution. Consequently, the normal distribution tends to underestimate the probability of the largest annual flood discharges.

Therefore, although the normal distribution provides a useful first approximation of the observed flood series, it does not fully reproduce the behaviour of the extreme upper tail. This limitation is particularly relevant when the distribution is later used for design flood calculations, where accurate representation of extreme discharges is important.

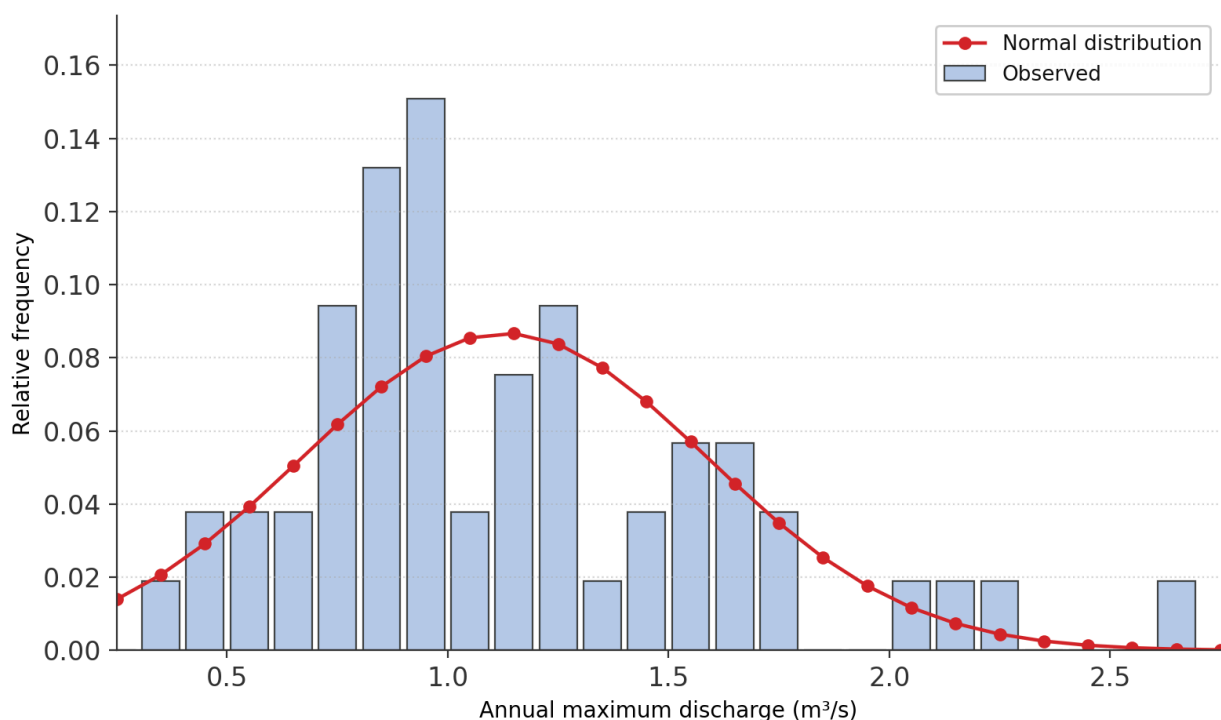


Figure 4. Comparison between the observed relative frequency distribution of annual maximum discharge and the relative frequencies predicted by the fitted normal distribution. The theoretical frequencies were calculated from the standard normal cumulative distribution using $0.1 \text{ m}^3/\text{s}$ discharge intervals.

Task 3 - Comparison Between Svarttjønnbekken and Tyholt

3.1 Methodology

Climate data from Svarttjønnbekken and the DNMI station at Tyholt were compared for the period 1973–1980. Annual precipitation was calculated by summing the available daily precipitation measurements for each year. For air temperature, the variable TAM (annual mean temperature) was used, as specified in the exercise.

The correspondence between the two stations was evaluated by comparing their annual values and calculating the Pearson correlation coefficient. A high correlation coefficient indicates that the two stations exhibit similar year-to-year variations, although it does not necessarily imply identical absolute values.

3.2 Results and Discussion

The comparison shows a strong correspondence between Svarttjønnbekken and Tyholt for both precipitation and temperature during 1973–1980.

For precipitation, the Pearson correlation coefficient is approximately:

$$r = 0.975$$

indicating a very strong positive correlation. The two stations therefore show very similar interannual variations in precipitation. However, the absolute precipitation amounts differ considerably. Svarttjønnbekken receives substantially more precipitation, with an average annual precipitation of approximately 1212 mm/year, compared with approximately 842 mm/year at Tyholt. This corresponds to an average difference of about 370 mm/year.

The temperature data also show a strong correspondence, with:

$$r = 0.945$$

This indicates that warm and cold years at Tyholt generally correspond to warm and cold years at Svarttjønnbekken. Nevertheless, Svarttjønnbekken is consistently colder, with an average annual mean temperature of approximately 3.01 °C, compared with 4.77 °C at Tyholt. The average difference is therefore approximately 1.76 °C.

Overall, the high correlation coefficients demonstrate that the two stations respond similarly to the regional climatic variability during 1973–1980. The systematic differences in mean precipitation and temperature are likely related to the different locations and local environmental conditions of the stations. Therefore, while Tyholt provides a good

representation of the temporal variability observed at Svarttjønnbekken, its measurements cannot be directly substituted for those from Svarttjønnbekken without accounting for the systematic differences in magnitude.

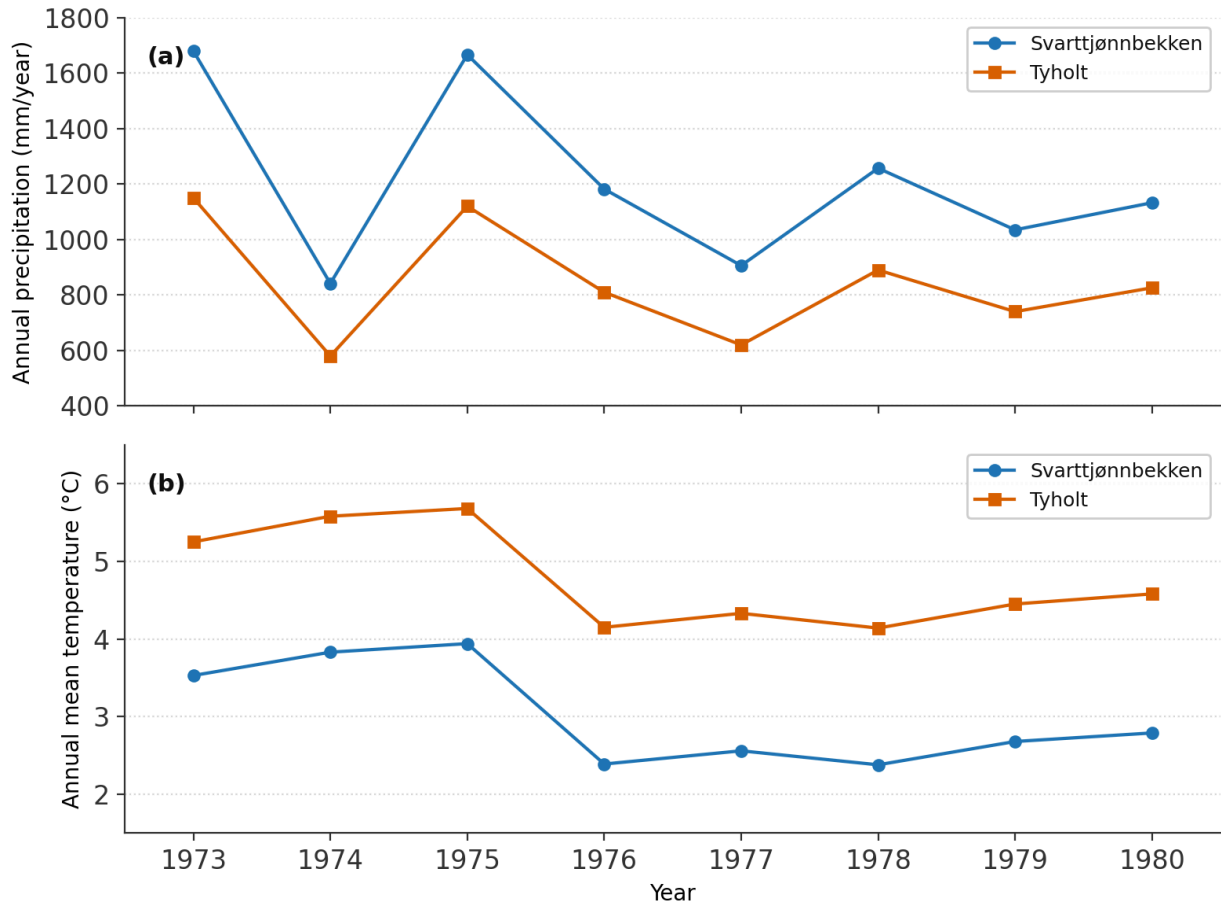


Figure 5. Comparison of annual precipitation and annual mean air temperature (TAM) between Svarttjønnbekken and Tyholt for the period 1973–1980. The two stations show strong temporal correspondence, with Pearson correlation coefficients of $r=0.975$ for precipitation and $r=0.945$ for temperature. Svarttjønnbekken is generally wetter and colder than Tyholt.

Task 4 - Linear Regression Between Tyholt and Svarttjønnbekken

4.1 Methodology

A linear regression was performed to establish a relationship between air temperature at the Tyholt station and temperature at Svarttjønnbekken. Following the instructions in

the exercise, Tyholt temperature was used as the independent variable (x), while Svarttjønnbekken temperature was used as the dependent variable (y). The regression was performed using paired daily temperature observations for the common period 1973–1980, yielding a total of 2920 data pairs.

4.2 Results and Discussion

The resulting regression equation is:

$$T_{Svarttjønnbekken} = 1.0408 T_{Tyholt} - 1.9389$$

The coefficient of determination is:

$$R^2 = 0.952$$

This indicates a strong linear relationship between the temperatures measured at the two stations. Approximately 95.2% of the variability in Svarttjønnbekken temperature is explained by the linear relationship with Tyholt temperature over the 1973–1980 period.

Using the regression equation, the estimated Svarttjønnbekken temperatures for the three specified Tyholt temperatures are:

Tyholt temperature (°C)	Estimated Svarttjønnbekken temperature (°C)
−3	−5.06
0	−1.94
+3	+1.18

For example, when Tyholt records 0 °C:

$$T_{Svarttjønnbekken} = 1.0408(0) - 1.9389 = -1.9389\text{ °C} \approx -1.94\text{ °C}$$

The regression therefore indicates that Svarttjønnbekken is generally colder than Tyholt, while temperature variations at the two stations are closely related. This relationship can subsequently be used to estimate missing Svarttjønnbekken temperature observations from corresponding Tyholt measurements.

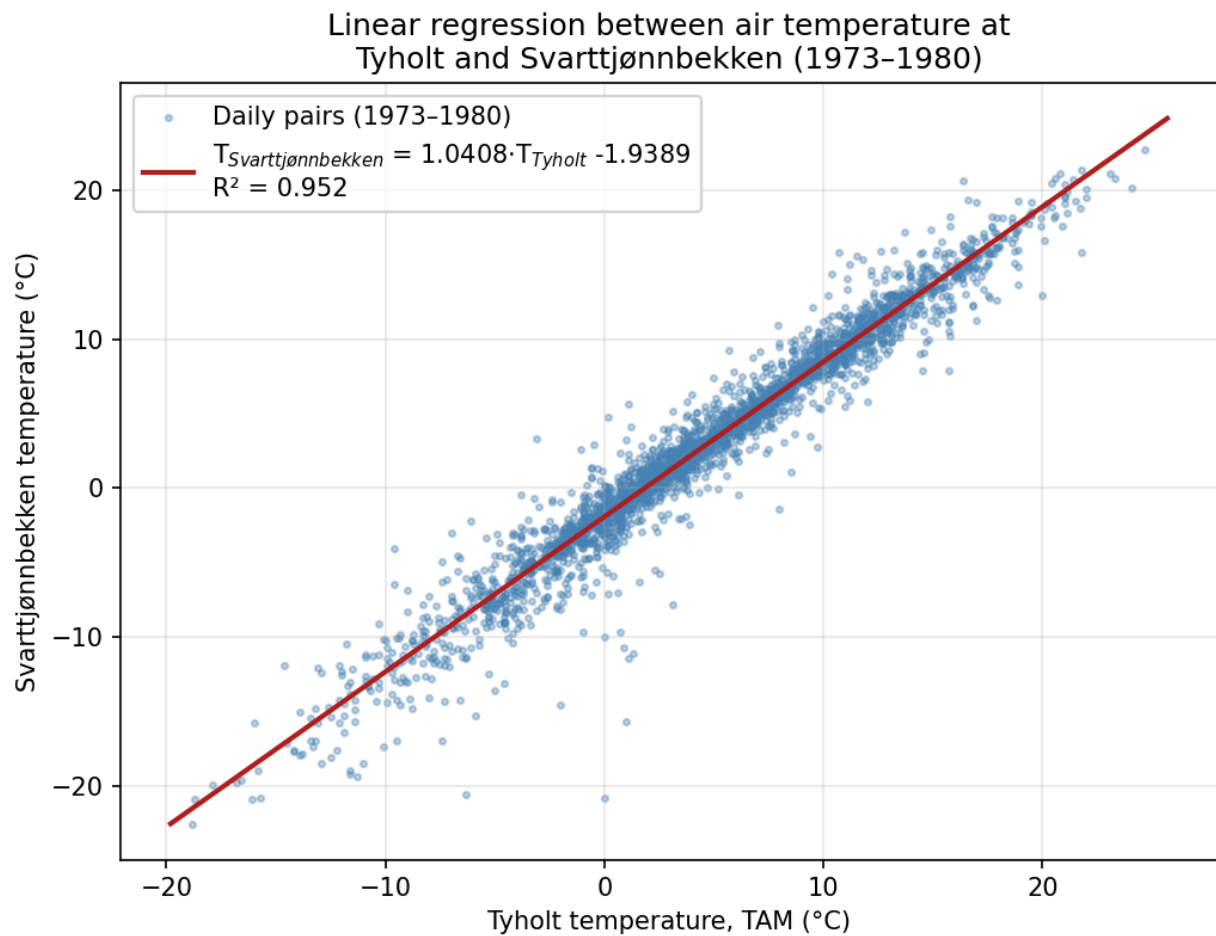


Figure 6. Linear regression between air temperature at Tyholt and Svarttjønnbekken for 1973–1980. Tyholt temperature is plotted on the x-axis and Svarttjønnbekken temperature on the y-axis. The fitted regression equation is $T_{Svarttjønnbekken} = 1.0192 T_{Tyholt} - 1.8528$, with $R^2 = 0.908$.

Task 5 - Flow Duration and Threshold Exceedance

5.1 Methodology

Daily discharge data from the Svarttjønnbekken gauging station for the last five years of record (2021–2025) were used to address this task, as suggested in the exercise. For

each year, the number of days on which the daily discharge exceeded $0.1 \text{ m}^3/\text{s}$ was counted.

In addition, the daily discharge values from these five years were combined into a single dataset (1825 daily observations) in order to construct a flow duration curve. The data were sorted in descending order, and the percentage of time each discharge value is equalled or exceeded was calculated as:

$$P_i = \frac{i}{n} \times 100$$

where i is the rank of the observation when sorted in descending order and $n = 1825$ is the total number of daily observations in the five-year period.

5.2 Results

The number of days per year on which discharge exceeded $0.1 \text{ m}^3/\text{s}$ is summarised in Table 5.

Table 5. Number of days per year with discharge exceeding $0.1 \text{ m}^3/\text{s}$ at Svarttjønnbekken (2021–2025).

Year	Days > $0.1 \text{ m}^3/\text{s}$	% of year
2021	115	31.5%
2022	106	29.0%
2023	104	28.5%
2024	116	31.8%
2025	142	38.9%
Average	116.6	31.9%

On average, discharge at Svarttjønnbekken exceeds $0.1 \text{ m}^3/\text{s}$ on approximately 117 days per year (about 32% of the year), although with noticeable interannual variation, ranging from 104 days in 2023 to 142 days in 2025.

The flow duration curve constructed from the combined 2021–2025 dataset is shown in Figure 7.

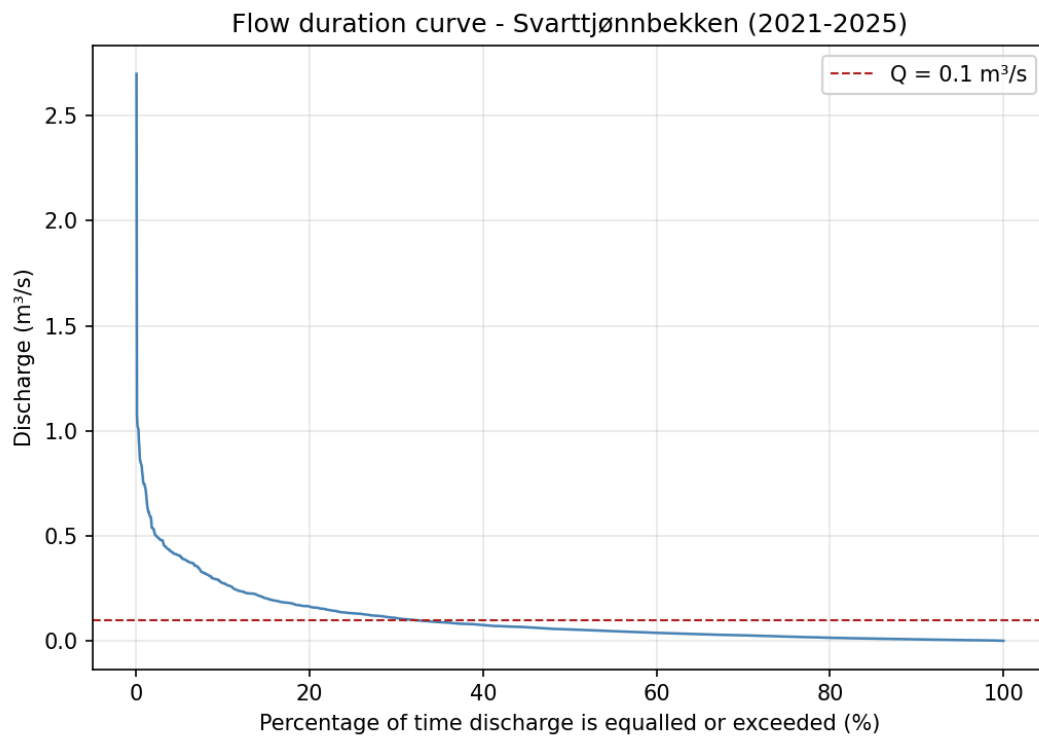


Figure 7. Flow duration curve for Svarttjønnbekken, based on daily discharge data for the period 2021–2025. The dashed red line indicates the $0.1 \text{ m}^3/\text{s}$ threshold.

The flow duration curve confirms this result: a discharge of $0.1 \text{ m}^3/\text{s}$ corresponds to an exceedance percentile of approximately 32%, meaning this discharge is equalled or exceeded roughly 32% of the time. The curve shows a steep decline at the higher discharges (the first 0–10% of time), reflecting occasional short-lived flood events, followed by a progressively flatter section at lower discharges, which is typical of a flow regime dominated by discrete precipitation and snowmelt events with relatively rapid recession.

5.3 Discussion

The results indicate that Svarttjønnbekken exceeds a discharge of $0.1 \text{ m}^3/\text{s}$ during just under one third of the year ($\sim 117 \text{ days}$), while for the remaining $\sim 68\%$ of the time discharge remains below this value. This is consistent with the characteristics of the catchment, which is small (3.04 km^2) and fast-responding: high discharges are linked to specific precipitation and snowmelt events, whereas baseflow conditions dominate for most of the year. Notably, 2025 stands out with the highest number of exceedance days (142 days , 38.9%), suggesting wetter conditions or a higher frequency of precipitation events in that year compared with the other four.

Task 6 - Long-Term Temperature Trend at Svarttjønnbekken

6.1 Methodology

To investigate a possible long-term trend in air temperature, the full annual mean temperature series for Svarttjønnbekken (1973–2024) was analysed. As noted in Section 1.1, the year 2005 contains no temperature observations and was therefore excluded from the calculations, leaving 51 years of annual mean values.

A temperature anomaly approach was used, following the procedure suggested in the exercise. First, the 30-year normal temperature was calculated as the arithmetic mean of the annual mean temperatures over the period 1991–2020:

$$T_{normal} = \frac{1}{30} \sum_{i=1991}^{2020} T_{annual,i}$$

The annual temperature anomaly for each year was then obtained by subtracting this normal value from the corresponding annual mean temperature:

$$\Delta T_i = T_{annual,i} - T_{normal}$$

A linear trend line was fitted to the resulting anomaly series using least-squares regression, in the same way a trendline would be added in Excel.

6.2 Results

The 1991–2020 normal temperature at Svarttjønnbekken is:

$$T_{normal} = 4.60^{\circ}C$$

The resulting anomaly series, together with the fitted linear trend, is shown in Figure 8.

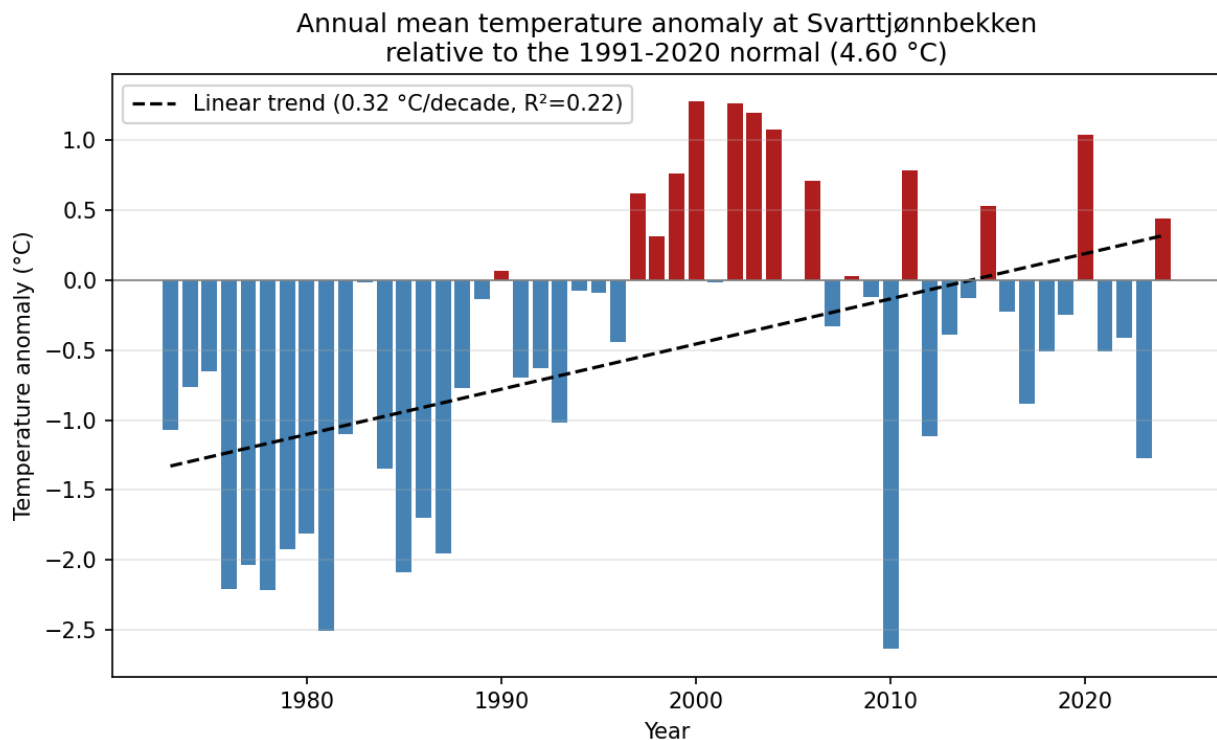


Figure 8. Annual mean temperature anomaly at Svarttjønnbekken (1973–2024) relative to the 1991–2020 normal of 4.60 °C. The dashed line shows the fitted linear trend. Year 2005 is excluded due to missing data.

The fitted trend line has a slope of:

$$0.0323 \text{ } ^\circ\text{C yr}^{-1} \approx 0.32 \text{ } ^\circ\text{C per decade}$$

with a coefficient of determination of:

$$R^2 = 0.22$$

The most negative anomaly occurred in 2010 ($-2.64 \text{ } ^\circ\text{C}$ relative to the normal), consistent with the exceptionally cold annual mean temperature identified in Section 1.3 ($1.96 \text{ } ^\circ\text{C}$). The most positive anomaly occurred in 2000 ($+1.27 \text{ } ^\circ\text{C}$), corresponding to the warmest year in the record ($5.87 \text{ } ^\circ\text{C}$).

6.3 Discussion

The anomaly plot shows a clear pattern: nearly all years before the mid-1990s have negative anomalies (i.e., colder than the 1991–2020 normal), while most years from the

late 1990s onward have positive anomalies. This shift is reflected in the positive slope of the fitted trend, corresponding to a warming of approximately 0.32 °C per decade, or roughly 1.6 °C over the full 1973–2024 period.

The R^2 value of 0.22 indicates that the linear trend explains only a modest proportion of the year-to-year variability, which remains substantial (individual anomalies range from about -2.6 °C to +1.3 °C). Nevertheless, the systematic shift from predominantly negative to predominantly positive anomalies over the five-decade record provides visual and statistical evidence of a long-term warming trend at Svarttjønnbekken, consistent with the broader pattern of regional and global climate change. Unlike the annual mean and annual maximum discharge series (Exercise 1, Task 2), which showed no discernible trend, the temperature record therefore does show a discernible, though moderate, long-term signal.

